

Practical Workbook
CT-361
Artificial Intelligence & Expert Systems



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LAB # 03

INPUT & OUTPUT COMMANDS

NOTE: The last question was skipped as instructed during the lab explanation.

Q1. Greeting by Name

- Write a Prolog program that:
- Prompts the user to enter their name.
- Reads the input name.
- Prints a greeting message using the entered name, like:
Hello, [Name]! Welcome to Prolog.

SOLUTION:

```
% Q1
greet:-
    write('Enter your name:'),
    read(Name),
    write('Hello, '),write(Name),write('!Welcome to Prolog. '),nl.
```

OUTPUT:

```
152 ?- greet.
Enter your name:moazzam.
Hello,moazzam!Welcome to Prolog.
true.
```

Q2. Simple Calculator

- Read two numbers and an operator (+, -, *, /)
- Perform the operation
- Print the result

SOLUTION:

```
% Q2
calculator :-
    write('Enter first number: '),read(N1),
    write('Enter second number: '),read(N2),
    write('Enter operator (+, -, *, /): '),read(Op),
    calculate(N1,N2,Op,Result),
    write('Result: '),write(Result),nl.

calculate(N1,N2,+,Result) :- Result is N1+N2.
calculate(N1,N2,-,Result) :- Result is N1-N2.
calculate(N1,N2,*,Result) :- Result is N1*N2.
calculate(N1,N2,/,Result) :- Result is N1/N2.
```

OUTPUT:

```
154 ?- calculator.
Enter first number: 6.
Enter second number: |: 4.
Enter operator (+, -, *, /): |: +
|: .
Result: 10
true .
```

```
154 ?- calculator.
Enter first number: 7.
Enter second number: |: 4.
Enter operator (+, -, *, /): |: -
|: .
Result: 3
true .
```

```
154 ?- calculator.
Enter first number: 2.
Enter second number: |: 3.
Enter operator (+, -, *, /): |: *
|: .
Result: 6
true .
```

```
154 ?- calculator.
Enter first number: 4.
Enter second number: |: 2.
Enter operator (+, -, *, /): |: /
|: .
Result: 2
true .
```

Q3. Check Even or Odd

Write a Prolog program that:

- Reads a number from the user.
- Checks if it is even or odd.
- Prints "The number [Number] is even." or "The number [Number] is odd." accordingly.

SOLUTION:

```
% Q3
check_even_odd :-
    write('Enter a number:'),
    read(Number),
    (Number mod 2 == 0 ->
        write(Number),write('is even.')
    ;
        write(Number),write('is odd.')).
```

OUTPUT:

```
159 ?- check_even_odd.
Enter a number:4.
4 is even.
true.
```

```
159 ?- check_even_odd.
Enter a number:5.
5 is odd.
true.
```

Q4. Calculate each using Prolog: Predict the output before running, then verify it in Prolog

1. X is $3 + 4 * 2$.
2. X is $(3 + 4) * 2$.
3. X is $10 * (5 + 2) - 8$.
4. Y = 2, X = 4, Find $Y * 5 + 10 * X$.
5. A = 3, B = 2, C is $(A + B) * (A - B)$.

PREDICTED OUTPUT:

1. 11
2. 14
3. 62
4. 50
5. 5

SOLUTION:

```
% Q4
calc1(X):-X is 3+4*2.
calc2(X):-X is (3+4)*2.
calc3(X):-X is 10*(5+2)-8.
calc4:-
    write('Enter Y: '),read(Y),
    write('Enter X: '),read(X),
    Ans is Y*5+10*X,
    write('Result: '),write(Ans),nl.
calc5:-
    write('Enter A: '),read(A),
    write('Enter B: '),read(B),
    C is (A+B)*(A-B),
    write('Result: '),write(C),nl.
```

VERIFICATION OF OUTPUT:

```
160 ?- calc1(X).  
X = 11.
```

```
161 ?- calc2(X).  
X = 14.
```

```
162 ?- calc3(X).  
X = 62.
```

```
163 ?- calc4.  
Enter Y: 2.  
Enter X: |: 4.  
Result: 50  
true.
```

```
164 ?- calc5.  
Enter A: 3.  
Enter B: |: 2.  
Result: 5  
true.
```